	Academic Semester	Fall 2026-2027	**Academic level**	1
	Course Code	MA104	**Exam**	Quiz 1 — Model B
	Course Title	Discrete Mathematics		
	Time Allowed	30 Minutes	**Total Points**	10
	Course Instructors	Dr. Salah Moawad , Dr. Tamer Seuodi		

Student Name: _____ **Student ID:** _____ **Section:** _____

*Answer **all** questions. Use T / F in truth tables and 1 / 0 in membership tables.*

Question 1 — Choose the correct answer and give the reason (5 points)

One point each: circle the letter of the correct answer **and** write a short reason in the box.

1. If p is **F** and q is **T**, which proposition is **true**?

A) $p \to q$ **B)** $p \wedge q$
C) $p \leftrightarrow q$ **D)** $\neg q$

Reason:

2. Let $A = \{2, 4, 6, 8\}$ and $B = \{4, 5, 6\}$. Then $A \cap B =$

A) $\{4, 6\}$ **B)** $\{2, 8\}$
C) $\{2, 4, 5, 6, 8\}$ **D)** $\{5\}$

Reason:

3. $\neg(p \wedge q) \equiv$?

A) $\neg p \wedge \neg q$ **B)** $\neg p \wedge q$
C) $p \vee \neg q$ **D)** $\neg p \vee \neg q$

Reason:

4. Is $((p \wedge q) \to r$ a **well-formed formula** (WFF)? If not, name the rule it breaks.

A) WFF **B)** Not a WFF

Reason:

5. Let $U = \{1, 2, 3, 4, 5, 6\}$, $A = \{1, 2, 5\}$, $B = \{2, 3, 5, 6\}$. Then $A^c \cap B =$

A) $\{3, 6\}$ **B)** $\{2, 5\}$
C) $\{4\}$ **D)** $\{1, 3, 4, 6\}$

Reason:

Question 2 — Truth table (2.5 points)

(a) Complete the truth table of $((p \to q) \wedge \neg q) \to \neg p$. [2 pts]

p	q	$p \to q$	$\neg q$	$\neg p$	$(p \to q) \wedge \neg q$	$((p \to q) \wedge \neg q) \to \neg p$
T	T					
T	F					
F	T					
F	F					

(b) Let P be the proposition in (a). Is $P \equiv p \vee q$? Justify using your table. [½ pt]

Question 3 — Membership table (2.5 points)

(a) Complete the membership table for the identity $A - (A \cap B) = A - B$. [2 pts]

A	B	$A \cap B$	$A - (A \cap B)$	$A - B$
1	1			
1	0			
0	1			
0	0			

(b) Is the identity $A - (A \cap B) = A - B$ **true** or **false**? Justify using your table. [½ pt]

— End of quiz · Good luck —